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MATHEMATICS **9709/13**

Paper 1 Pure Mathematics 1 **October/November 2025**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

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Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
FT	Follow through
~~	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
J	Judgement
JU	Judgement
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two
MR	Misread
O	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
PE	Judgment made by the PE
Pre	Premature approximation
SC	Special case
SEEN	Indicates that work/page has been seen

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Annotation	Meaning
SF	Error in number of significant figures
	Correct
TE	Transcription error
XP	Correct answer from incorrect working

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	$\left[\binom{6}{0}\right]2^6 \pm \binom{6}{1}2^5\left(\frac{x}{2}\right) \pm \binom{6}{2}2^4\left(\frac{x}{2}\right)^2 \pm \binom{6}{3}2^3\left(\frac{x}{2}\right)^3$ or $2^6\left(1 \pm 6\left(\frac{x}{4}\right) \pm \frac{6.5}{2!}\left(\frac{x}{4}\right)^2 \pm \frac{6.5.4}{3!}\left(\frac{x}{4}\right)^3\right)$	**M1**	OE A correct unsimplified expansion of at least four terms. Condone $\pm$ sign errors for this mark. Condone omission of brackets if recovered. This mark can be implied by sight of $64+96x+60x^2+20x^3$.
	$64-96x+60x^2-20x^3$	**A1**	Three correct simplified terms.
		A1	Four correct simplified terms. Accept these terms given in a list.
		3	
1(b)	$(3\times \textit{their}(-20))+(-1\times \textit{their}\,60)+(2\times \textit{their}\,64)$	**M1**	Selecting the correct products using *their* expansion.
	8	**A1**	Accept $8x^3$.
		2	

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Question	Answer	Marks	Guidance
2(a)	$5x-3=\tan\left(-\frac{1}{4}\pi\right)$ or $-\tan\left(\frac{1}{4}\pi\right)[=-1]$	**M1**	Condone $5x-3=\tan\left(\frac{1}{4}\pi\right)$ and $5x-3=-0.0137\ldots$
	$x=\frac{2}{5}$ or 0.4	**A1**	
		2	
2(b)	$5-5\sin^2\theta=4\sin\theta+4$	***M1**	Replacing $\cos^2\theta$ with $1-\sin^2\theta$.
	$5\sin^2\theta+4\sin\theta-1[=0]$ $\quad[\Rightarrow(5\sin\theta-1)(\sin\theta+1)=0]$	**DM1**	Collecting terms to form a three-term quadratic. Condone only ± errors.
	$\sin\theta=\frac{1}{5}$ and -1	**A1**	
	$\theta=2.94,\ 4.71$	**A1**	AWRT Accept $\left(\frac{3}{2}\pi\right)$ in place of 4.71 and 0.936π or $\left(\frac{337}{360}\pi\right)$ in place of 2.94. Condone repetition of $\frac{3}{2}\pi$ but no other values within the range. Ignore answers outside the range $\frac{\pi}{2}\leqslant\theta\leqslant 2\pi$.
		4	

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Question	Answer	Marks	Guidance
3(a)	72.030004	**B1**	CAO. Not AWRT.
		1	
3(b)	30.3888	**B1**	CAO. Not AWRT.
		1	
3(c)	30[.0]	**B1**	CAO. 30 may be accompanied by ‘around’, ‘approximately’ etc.
		1	

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Question	Answer	Marks	Guidance
4(a)	$20+(30-1)\times(their-5)$	**M1**	Correct expression for u_{30} using *their* calculated value of d.
	-125	**A1**	
		2	
4(b)	$r=\frac{k}{20}$ and $r=\frac{k-5}{k}$	**B1**	OE For two correct expressions for r. Can be implied by $\frac{k}{20}=\frac{k-5}{k}$ OE.
	$k^2-20k+100[=0]$	**M1**	Forming a 3-term quadratic. Condone only ± errors.
	$[r=]\frac{1}{2}$	**A1**	May be seen in the S_∞ formula.
	$\left[S_\infty=\frac{20}{1-\frac{1}{2}}=\right]40$	**A1**	
	Alternative Method for Question 4(b)		
	$20r=k$ and $20r^2=k-5$	**B1**	Stating two correct relationships between k and r. This can be implied by $20r^2=20r-5$ OE.
	$20r^2-20r+5[=0]$	**M1**	Forming a three-term quadratic in r. Condone only ± errors.
	$[r=]\frac{1}{2}$	**A1**	May be seen in the S_∞ formula.
	$\left[S_\infty=\frac{20}{1-\frac{1}{2}}=\right]40$	**A1**	
		4	

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Question	Answer	Marks	Guidance
5(a)	$\sin\left(\frac{1}{2}\theta\right)=\frac{2\sqrt{3}}{4}$ or $\cos\left(\frac{1}{2}\theta\right)=\frac{2}{4}$ or $\cos\theta=\frac{4^2+4^2-\left(4\sqrt{3}\right)^2}{2\times4\times4}\left[=-\frac{1}{2}\right]$	**B1**	Use of a correct expression for either $\frac{\theta}{2}$ or θ.
	$[\theta=]\frac{2\pi}{3}$	**B1**	Correct answer can imply B1B1. Condone inconsistent use of θ.
		2	
5(b)	$\frac{1}{2}\times4^2\times\sin\frac{2\pi}{3}$ *or* $\frac{1}{2}\times4^2\times\frac{4\pi}{3}$ *or* $\frac{1}{2}\times4^2\times\frac{2\pi}{3}$	***M1**	Use of a correct formula with *their* θ or $2\pi-$ *their* $\theta, \theta>0$. Triangle and/or minor sector may be seen in the formula for finding the minor segment.
	$\frac{1}{2}\times4^2\times\frac{4\pi}{3}+\frac{1}{2}\times4^2\times\sin\frac{2\pi}{3}$ Or $16\pi-\left(\frac{1}{2}\times4^2\times\frac{2\pi}{3}-\frac{1}{2}\times4^2\times\sin\frac{2\pi}{3}\right)$	**DM1**	Addition of relevant sector area with $2\pi-$ *their* θ, where $\theta>0$, and triangle area with *their* θ, where $\theta>0$. Subtraction of triangle area, with *their* $\theta, \theta>0$, from relevant sector area, with *their* $\theta, \theta>0$, and subtract from circle area.
	$\frac{32\pi}{3}+4\sqrt{3}$	**A1**	OE. Exact equivalents only. **SC B2** following M0DM0 for 40.4 AWRT.
		3	

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Question	Answer	Marks	Guidance
6(a)	{Stretch} {[scale] factor 2} {[parallel to] in/on y[-axis] or x-axis invariant}	**B2,1,0**	B2 for all three {} elements correct. B1 for two {} elements correct.
	{Translation} $\begin{pmatrix}\{3\}\\\{-4\}\end{pmatrix}$ or {[+]3 in/on [the] x[direction]} and {–4 in/on [the] y [direction]}	**B2,1,0**	B2 for all three {} elements correct. B1 for two {} elements correct. Do not condone use of 'Right' and 'Down' in place of x and y directions. Note: The vectors can be seen as $\begin{pmatrix}3\\0\end{pmatrix}$ and $\begin{pmatrix}0\\-4\end{pmatrix}$.
	Correct transformations in the correct order without any extra transformations. Down and right can be condoned for this mark if the candidate's intention is clear.	**B1**	Combined translations (condone transformations accompanied by a vector) must come after the stretch or, if separate, the $\begin{pmatrix}0\\-4\end{pmatrix}$ must come after the stretch.
	Alternative Method for Question 6(a)		
	{Translation} $\begin{pmatrix}\{3\}\\\{-2\}\end{pmatrix}$ or {[+]3 in/on [the] x [direction]} and {-2 in/on [the] y [direction]}	**B2,1,0**	B2 for all three {} elements correct. B1 for two {} elements correct. Do not condone use of 'Right' and 'Down' in place of x and y directions. Note: The vectors can be seen as $\begin{pmatrix}3\\0\end{pmatrix}$ and $\begin{pmatrix}0\\-2\end{pmatrix}$.
	{Stretch} {[scale]factor 2} {[parallel to] in/on y-[axis] or x axis invariant OE}	**B2,1,0**	
	Numerically correct transformations in the correct order without any extra transformations. Down and right can be condoned for this mark if the candidate's intention is clear.	**B1**	Combined translations (condone transformations accompanied by a vector) must come before the stretch or, if separate, the $\begin{pmatrix}0\\-2\end{pmatrix}$ must come before the stretch.
		5	

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Question	Answer	Marks	Guidance
6(b)	$a=2$	**B1**	Individual values are considered the final answer and NOT those in $a(x+b)^3+c$, unless individual values are not stated.
	$b=-3$	**B1**	
	$c=-4$	**B1**	
		3	

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Question	Answer	Marks	Guidance
7	$\left[y=\frac{2}{ax-3}+\frac{1}{2}\Rightarrow\right]\ \left(y-\frac{1}{2}\right)(ax-3)=2$ or $ax-3=\frac{2}{y-\frac{1}{2}}$	*M1	OE For setting '$y=$' and rearranging. Condone only ± errors.
	$ax=\frac{4}{2y-1}+3$	DM1	For making ax the subject. Condone only ± errors.
	$[y=$ or $g^{-1}(x)=\ ]\frac{4}{a(2x-1)}+\frac{3}{a}$ or $\frac{2}{a\left(x-\frac{1}{2}\right)}+\frac{3}{a}$	A1	OE Note: x and y may be interchanged from the start.
	Clear replacement of a with 6 in g^{-1} and, if necessary, followed by simplification to arrive at exactly the same expressions.	A1	This mark is dependent on getting the correct inverse.
	Alternative Method for Question 7		
	$\left[y=\frac{ax+1}{2(ax-3)}\Rightarrow\right]2y(ax-3)=ax+1$	*M1	OE For setting '$y=$' and rearranging. Condone only ± errors.
	$ax(2y-1)=6y+1$	DM1	For making $ax(2y-1)$ the subject, condone only ± errors.
	$[y=$ or $g^{-1}(x)=\ ]\frac{6x+1}{a(2x-1)}$ or $\frac{3x+\frac{1}{2}}{a\left(x-\frac{1}{2}\right)}$	A1	OE Note: x and y may be interchanged from the start.
	Clear replacement of a with 6 in g^{-1} and, if necessary, followed by simplification to arrive at exactly the same expressions.	A1	This mark is dependent on getting the correct inverse.
		4	

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Question	Answer	Marks	Guidance
8(a)	Centre is $(2, 4)$	**B1**	
	[(Radius)^2 =] $(6-2)^2+(1-4)^2$ or $\left(\frac{1}{2}\sqrt{(6-(-2))^2+(1-7)^2}\right)^2$ [$=25$]	**M1**	Use of a correct method to find $(\text{radius})^2$ or radius. This can be implied by $r=5$.
	$(x-2)^2+(y-4)^2=(5)^2$	**A1**	OE
	Alternative Method for Question 8(a)		
	Either $\frac{y-1}{x-6}$ or $\frac{y-7}{x+2}$	**B1**	Correct expression using either $(6,1)$ or $(-2,7)$ and (x,y).
	$\frac{y-1}{x-6}\times\frac{y-7}{x+2}=-1$	**M1**	Product of *their* gradients $=-1$.
	$x^2+y^2-4x-8y-5=0$	**A1**	OE Simplification to get the correct five terms.
		3	

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Question	Answer	Marks	Guidance
8(b)	[Gradient of normal through centre =] 2	**B1**	SOI
	$y-4=(their\ gradient)(x-2)$ Or $4=(their\ 2)(2)+c \Rightarrow c=$ [0]	***M1**	Correct form of a line equation with *their* normal gradient i.e. not $-\frac{1}{2}$
	$(x-2)^2+(2x-4)^2=25$	**DM1**	OE Using $their\ 2x$ and *their* circle to form an equation in x.
	$\left[5(x-2)^2=25 \Rightarrow\right](x-2)^2=5$	**A1**	OE Accept $x^2-4x-1[=0]$.
	$[x=]2\pm\sqrt{5}$	**A1**	OE Exact values only.
	Alternative Method for Question 8(b)		
	$2x+2y\frac{\mathrm{d}y}{\mathrm{d}x}-4-8\frac{\mathrm{d}y}{\mathrm{d}x}=0$	**B1**	OE Implicit differentiation of the correct circle equation.
	$2x+2y\left(-\frac{1}{2}\right)-4-8\left(-\frac{1}{2}\right)=0$	***M1**	Replacing $\frac{\mathrm{d}y}{\mathrm{d}x}$ with $-\frac{1}{2}$ in their differential which must contain two terms in $\frac{\mathrm{d}y}{\mathrm{d}x}$. Expect $y=2x$.
	$(x-2)^2+(2x-4)^2=25$	**DM1**	OE Using *their* $2x$ and *their* circle to form an equation in x.
	$\left[5(x-2)^2=25 \Rightarrow\right](x-2)^2=5$	**A1**	OE Accept $x^2-4x-1[=0]$.
	$[x=]2\pm\sqrt{5}$	**A1**	OE Exact values only.

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Question	Answer	Marks	Guidance
8(b)	**Alternative Method 2 for Question 8(b)**		
	$y=\sqrt{25-(x-2)^2}+4$ or $\left(-x^2+4x+21\right)^{\frac{1}{2}}+4$	**B1**	This mark can be implied by a correct differential.
	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2}\left(25-(x-2)^2\right)^{-\frac{1}{2}}\times(4-2x)$	***M1**	Differentiating *their* expression for y. Use of the chain rule must be clear.
	$-\frac{1}{2}=\frac{1}{2}\left(25-(x-2)^2\right)^{-\frac{1}{2}}\times(4-2x)\quad\left[\Rightarrow 25-(x-2)^2=(2x-4)^2\right]$	**DM1**	Equating *their* $\frac{\mathrm{d}y}{\mathrm{d}x}$ to $-\frac{1}{2}$.
	$5x^2-20x-5=0$	**A1**	OE
	$[x=]2\pm\sqrt{5}$	**A1**	OE Exact values only.
	Alternative Method 3 for Question 8(b)		
	$(x-2)^2+\left(\left(-\frac{1}{2}x+c\right)-4\right)^2=25$	**B1**	Replacing y with $-\frac{1}{2}x+c$ in a correct circle equation.
	$\frac{5}{4}x^2-cx+(c-4)^2-21=0$ or $\frac{5}{4}x^2-cx+\left(c^2-8c-5\right)=0$	**M1**	Forming a three-term quadratic in x. Condone only ± errors.
	Using "b^2-4ac"$=0$ with *their* values from the quadratic in x	**DM1**	Expect $c^2-4\times\frac{5}{4}\times\left\{(c-4)^2-21\right\}=0$.
	$[c=]\frac{40\pm\sqrt{1600+400}}{8}\left[=\frac{10\pm5\sqrt{5}}{2}\right]$	**A1**	OE
	$[x=]2\pm\sqrt{5}$	**A1**	OE Exact values only.
		5	

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Question	Answer	Marks	Guidance
9	$\frac{1}{2}x+\frac{4}{x}=4.5\Rightarrow x^2-9x+8=0$	**M1**	Equating the curve and the line to form a 3-term quadratic.
	$[x=]1,\ 8$	**A1**	SOI
	$\pi\times4.5^2\times(\textit{their}\ 8-\textit{their}\ 1)\left[=\frac{567}{4}\pi \text{ or } 445.3\right]$	**A1 FT**	FT *their* limits from the quadratic, both limits must be positive.
	$[\pi]\int\left(\frac{1}{2}x+\frac{4}{x}\right)^2[\mathrm{d}x]$	***M1**	For use of $\int y^2\mathrm{d}x$.
	$[\pi]\int\left(\frac{1}{4}x^2+4+\frac{16}{x^2}\right)\mathrm{d}x=[\pi]\left(\frac{1}{12}x^3+4x-\frac{16}{x}\right)$	**A1**	OE First A1 for two correct terms, second A1 for all three correct. May be unsimplified. May include +*C*.
		A1	
	$[\pi]\left(\frac{218}{3}+\frac{143}{12}\right)=[\pi]\frac{1015}{12}$	**DM1**	Clear use of their *limits* in *their* integrated expression. At least one correct power required. Both limits must be positive, or if correct limits are used, then $\pm\left(\frac{218}{3}+\frac{143}{12}\right)$ is the minimum required.
	Volume of solid = $\frac{343}{6}\pi$	**A1**	Condone a negative answer changed to a positive. A0 for omission of π. **SC B1**: following DM0. A0 for 179.594…

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Question	Answer	Marks	Guidance
	Alternative Method for Question 9		
	$\frac{1}{2}x+\frac{4}{x}=4.5\Rightarrow x^2-9x+8=0$	**M1**	Equating the curve and the line to form a 3-term quadratic
	$[x=]1,\ 8$	**A1**	SOI
	[Volume =] $[\pi]\int\left(4.5^2-\left(\frac{1}{2}x+\frac{4}{x}\right)^2\right)[\mathrm{d}x]$	***M1**	For use of $\int\left(y_1{}^2-y_2{}^2\right)$. Condone $\int\left(y_1-y_2\right)^2$ for this *M1.
	$\pm[\pi]\left(\left\{4.5^2x-4x\right\}\left\{-\frac{1}{12}x^3\right\}\left\{+\frac{16}{x}\right\}\right)$	**A1** **A1** **A1**	OE A1 for each element of the integral. May be unsimplified. May include +*C*. 'Correct' terms from $\int\left(y_1-y_2\right)^2$ score A0A0A0 XP.
	$[\pi]\left(\frac{268}{3}-\frac{193}{6}\right)$	**DM1**	Clear use of *their* limits in their integrated expression. At least one correct power is required. Both limits must be positive, or if correct limits are used, then $\pm\left(\frac{268}{3}-\frac{193}{6}\right)$ is the minimum required.
	Volume of solid = $\frac{343}{6}\pi$	**A1**	Condone a negative answer changed to a positive. A0 for omission of π. **SC B1**: following DM0. A0 for 179.594…
		8	

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Question	Answer	Marks	Guidance
10(a)(i)	$\{2\}\{(x+1)^2\}\{+8\}$	**B1**	B1 for each {} element.
		B1	
		B1	
		3	
10(a)(ii)	$[\text{Range or f}[(x)] \text{ or } y] \geqslant 8$	**B1 FT**	OE. FT *their* value of c. $x \geqslant 8$ scores B0.
		1	

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Question	Answer	Marks	Guidance
10(b)	$p(5m)^2+4(5m)-5[=0]$ and $p(-9m)^2+4(-9m)-5[=0]$	*M1	Allow one error for this mark if the intention is clear.
	$81\left(\frac{5-20m}{(5m)^2}\right)m^2+4(-9m)-5[=0]$ or $25\left(\frac{36m+5}{(-9m)^2}\right)m^2+4(5m)-5[=0]$ or $\left(\frac{5-20m}{(5m)^2}\right)=\left(\frac{36m+5}{(-9m)^2}\right)$	*DM1	Eliminate p by substitution or equating two expressions for p.
	$2520m=280$	DM1	OE Simplify as far as a linear equation or a quadratic where $m=0$ is ignored or dismissed.
	$m=\frac{1}{9}$, $p=9$	A1, A1	
	Alternative Method for Question 10(b)		
	$p(5m)^2+4(5m)-5[=0]$ and $p(-9m)^2+4(-9m)-5[=0]$	*M1	Allow one error for this mark if the intention is clear.
	$p(5m)^2+4(5m)-5=p(-9m)^2+4(-9m)-5$	*DM1	Equating *their* two expressions.
	$\left[56pm^2-56m=0\Rightarrow m=\frac{1}{p}\Rightarrow\right]\ 5=\frac{45}{p}$ or $45m=5$	DM1	Simplify as far as a quadratic where $m=0$ is ignored or dismissed and substitute for m or p.
	$p=9$, $m=\frac{1}{9}$	A1, A1	

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Question	Answer	Marks	Guidance
10(b)	**Alternative Method 2 for Question 10(b)**		
	$\frac{-4+\sqrt{4^2-4(p)(-5)}}{2p}=5m$ and $\frac{-4-\sqrt{4^2-4(p)(-5)}}{2p}=-9m$	***M1**	Equating $5m$ and $-9m$ to the two solutions from the quadratic formula. Allow one error for this mark if the intention is clear.
	$\frac{-4+\sqrt{4^2-4(p)(-5)}}{(5\times 2)p}=\frac{-4-\sqrt{4^2-4(p)(-5)}}{(-9\times 2)p}$	***DM1**	Attempt to solve simultaneously by equating *their* expressions for m, or dividing one by the other and equating to $-\frac{5}{9}$.
	$16+20p=196$	**DM1**	OE Simplify as far as a linear expression – no square root.
	$p=9,\ m=\frac{1}{9}$	**A1, A1**	
	Alternative Method 3 for Question 10(b)		
	$\left[(x-5m)(x+9m)\Rightarrow\right]\quad x^2+4mx-45m^2\left[=0\right]$ Or $\alpha+\beta=-4m$ and $\alpha\beta=-45m^2$	***M1**	Recognising the relationship between $5m$ and $-9m$ and the quadratic equation.
	$4m=\frac{4}{p},\ -45m^2=-\frac{5}{p}$	***DM1**	Using $x^2+\frac{4}{p}x-\frac{5}{p}\left[=0\right]$. Comparing coefficients or using $\alpha+\beta=-\frac{b}{a}$ and $\alpha\beta=\frac{c}{a}$.
	$m=\frac{1}{p}\ \Rightarrow\ -45m^2=-5m$ or $p=\frac{1}{m}\Rightarrow\ 5p^2=45p$	**DM1**	Forming an equation in one unknown.
	$m=\frac{1}{9},\ p=9$	**A1, A1**	
		5	

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Question	Answer	Marks	Guidance
11(a)	$\frac{dy}{dx} = \left\{\frac{-8\times 3}{(3x-8)^2}\right\}\left\{+\frac{6}{(x-1)^2}\right\}$	**B1** **B1**	B1 for each {} element.
	$\left\{\frac{-8\times 3}{(3.3-8)^2}\right\}\left\{+\frac{6}{(3-1)^2}\right\}\left[=\frac{-45}{2}\right]$	***M1**	Using *their* differentiated expression, which must contain $(3x-8)^{-2}$ and $(x-1)^{-2}$, and $x=3$. This may be seen in *their* line equation.
	$y-5=\left(their\ \frac{-45}{2}\right)(x-3)$ or $5=\left(their\ \frac{-45}{2}\right)(3)+c \Rightarrow c=\left[\frac{145}{2}\right]$	***DM1**	Correct form of a line equation with their gradient, but not the negative reciprocal, and $(3,\ 5)$.
	$\left[-8x-5=\frac{-45}{2}(x-3)\Rightarrow\right]\ 29x=145$	**DM1**	Replacing y with $-8x$ and collecting terms.
	$x=5, y=-40$	**A1**	Accept $(5,\ -40)$.
		6	

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Question	Answer	Marks	Guidance
11(b)(i)	$\dfrac{-24}{(3x-8)^2}+\dfrac{6}{(x-1)^2}=0 \Rightarrow 24(x-1)^2=6(3x-8)^2$ $\left[\Rightarrow 4(x-1)^2=(3x-8)^2\right]$	***M1**	Equate *their* $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, which must contain $(3x-8)^{-2}$ and $(x-1)^{-2}$ to 0, and clear of fractions. Only condone ± errors.
	$5x^2-40x+60$ [= 0] or $\pm 2(x-1)=3x-8$	**DM1**	OE Forming a three-term quadratic. Condone only ± errors. Or taking square roots. For this method, the ± must be present.
	2, 6	**A1**	Both values.
		3	
11(b)(ii)	$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\left\{\dfrac{-24\times -2\times 3}{(3x-8)^3}\right\}+\left\{\dfrac{6\times -2}{(x-1)^3}\right\}$	**B1FT** **B1FT**	Correct differentials of the elements of *their* $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, which must contain $(3x-8)^{-2}$ and $(x-1)^{-2}$.
	At $x=2$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\dfrac{\textit{their}\,144}{(6-8)^3}-\dfrac{\textit{their}\,12}{(2-1)^3}$ and at $x=6$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\dfrac{\textit{their}\,144}{(18-8)^3}-\dfrac{\textit{their}\,12}{(6-1)^3}$	**M1**	Replacing x with *their* 2 and *their* 6 in *their* $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$, which must contain $(3x-8)^{-3}$ and $(x-1)^{-3}$. Correct final answers.
	At $x=2$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=-30<0$, therefore max[imum] At $x=6$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\dfrac{48}{10^3}\left[\dfrac{6}{125}\right]>0$, therefore min[imum]	**A1**	WWW Correct values, or working, and >0 and <0 are required.
		4	

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